

Visualization Draft

1. Set up

Packages

```
# general use (data wrangling and plotting)
library(tidyverse)

# clean column names
library(janitor)

# file organization
library(here)
```

Data

```
# weather data from NOAA weather station
weather <- read_csv(here("data", "NOAA-weather-data.csv"))

# water quality survey metadata
metadata <- read_csv(here("data", "NCOS_YSI_Water_Quality_Monitoring_0.csv"))

# water quality parameter measurements
water_parameters <- read_csv(here("data", "YSI_Data_Begin_1.csv"))
```

Cleaning weather

```

# creating new object from weather
weather_clean <- weather |>
  # cleaning column names
  clean_names() |>
  # making sure date is in POSIXct format
  mutate(date = mdy(date)) |>
  # selecting columns of interest
  select(date, prcp, tmax, tmin) |>
  # extracting month and year from date column
  mutate(month = month(date),
         year = year(date)) |>
  # assigning water year
  mutate(wy = case_when(
    # when month is October or later, assign value of year + 1
    month >= 10 ~ year + 1,
    # when month is September or earlier, assign value of existing year
    .default = year))

```

Cleaning metadata

```

# creating new object from metadata
metadata_clean <- metadata |>
  # cleaning column names
  clean_names() |>
  # excluding unnecessary columns
  select(!c(object_id, creator, editor)) |>
  # making sure monitoring_date is in POSIXct format
  mutate(monitoring_date = mdy_hms(monitoring_date)) |>
  # renaming monitoring datetime column to reflect values
  rename(monitoring_datetime = monitoring_date) |>
  # creating new columns for month and year of observations
  mutate(month = month(monitoring_datetime),
         year = year(monitoring_datetime)) |>
  # assigning water year
  mutate(wy = case_when(
    # when month is October or later, assign value of year + 1
    month >= 10 ~ year + 1,
    # when month is September or earlier, assign value of existing year
    .default = year)) |>
  # create new column of full site name

```

```
mutate(site_full = case_when(
  site_name == "east_channel" ~ "East Channel",
  site_name == "phelps_bridge" ~ "Phelps Bridge",
  site_name == "venoco_bridge" ~ "Venoco Bridge",
  site_name == "other" ~ "Other"
))
```

Cleaning water parameters

```
# creating new object from water_parameters
water_parameters_clean <- water_parameters |>
  # cleaning object names
  clean_names() |>
  # excluding unnecessary columns
  select(!c(creator, editor)) |>
  # renaming columns
  rename(elevation_code = depth_code) |>
  # replacing incorrect elevation code from 2021-07-16
  mutate(elevation_code = case_when(
    elevation_code == 10 ~ 1,
    .default = elevation_code
  )) |>
  # set factors
  mutate(elevation_code = as_factor(elevation_code),
    elevation_code = fct_relevel(elevation_code,
      "0", "1", "2", "3", "4", "5", "6"))
```

Joining data frames

```
# join metadata and water parameters
water_quality <- full_join(
  # left
  x = metadata_clean,
  # right
  y = water_parameters_clean,
  # shared column
  by = c("global_id" = "parent_global_id")) |>
  # stop and look at colnames(water_quality) before continuing
```

```
# selecting relevant columns
select(
  # information about observation
  global_id, monitoring_datetime, time_of_wq_measurement,
  monitor_name, weather, site_name, other_site_name, site_full, x, y,

  # water information
  water_surface_elevation, flow, notes,

  # measurement information
  elevation_code, do_mg_l, do_percent_sat, conductivity_u_s_cm,
  salinity_ppt, temperature_c) |>
# extract date from monitoring datetime
mutate(date = date(monitoring_datetime)) |>
# move date column to appear to the right of monitoring_datetime
relocate(date, .after = monitoring_datetime) |>
# exclude monitoring_datetime
select(!monitoring_datetime)
```

Figure 1

Plan

Presenting how dissolved oxygen (DO) differ across sites

- x-axis: different site names
- y-axis: DO
- geom_jitter: jitter to represent individual DO measurements
- red point: mean DO for each site

Figure 2

Plan

Presenting how dissolved oxygen (DO) varies across elevations at each site.

- x-axis: DO
- y-axis: Elevation code

- `geom_point`: visualize individual observations
- `geom_jitter`: jitter to represent individual DO measurements

Figure 3